COMPUTER NETWORKS

Submitted in partial fulfilment of the requirement of the degree of

BACHELORS OF TECHNOLOGY

CSE

To

K.R Mangalam University

by

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Introduction

This assignment examines the core network topologies commonly used in computer networks, including Star topology using a switch, Star topology using a hub, a Ring-type topology implemented with multiple switches and Spanning Tree Protocol (STP), and a failure analysis performed on the ring topology. All network configurations and tests were carried out using Cisco Packet Tracer. Connectivity between devices was verified through ping commands, and the results and behavior of each topology were carefully observed and documented.

Before analyzing each topology individually, it is essential to understand the difference between how hubs and switches forward data frames. During the initial transmission, both hubs and switches send the frame to all connected ports because they do not yet have information about the destination device. However, after this first transmission, a switch begins to build and use a MAC address table. This allows it to send future frames only through the specific port associated with the destination device, reducing unnecessary network traffic. In contrast, a hub does not have the ability to learn or store MAC addresses. As a result, it continues to broadcast every frame to all connected devices regardless of how many transmissions occur. Due to this intelligent forwarding capability, switches provide better efficiency and performance compared to hubs, especially in networks with multiple connected devices.

Task 1: Star Topology (Switch)

A star topology was built in Cisco Packet Tracer using a 2960-24TT switch (Switch1) as the central device, with four PCs (PC0–PC3) connected to it. All devices were assigned IP addresses in the **192.168.10.x/24** subnet. Ping tests were performed from PC0 to each of the remaining PCs.

Network Topology Screenshots:

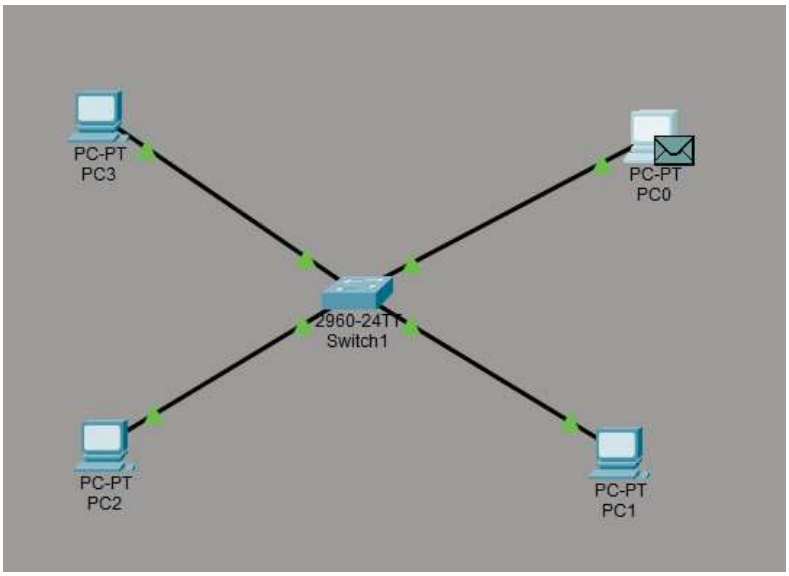

Fig 1a: PC0 initiates ping — hub broadcasts frame to all ports (first transmission)

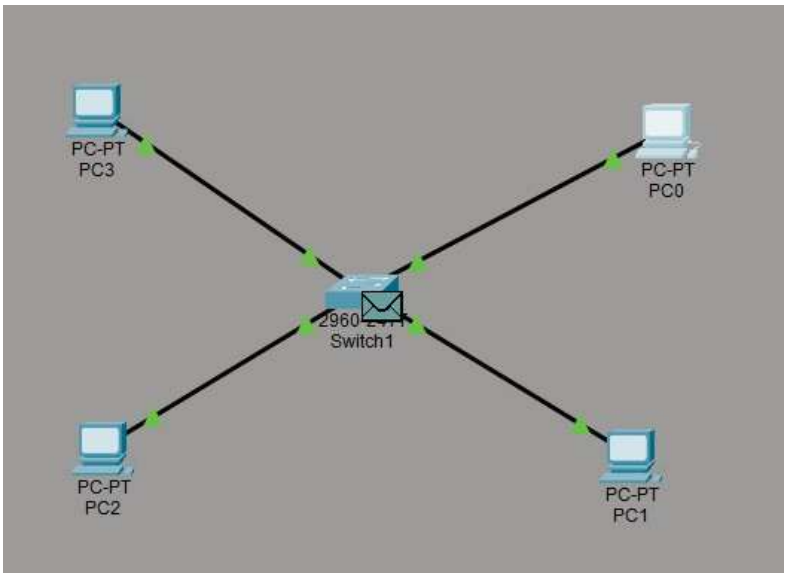

Fig 1b: Frame forwarded — switch learning MAC addresses in progress

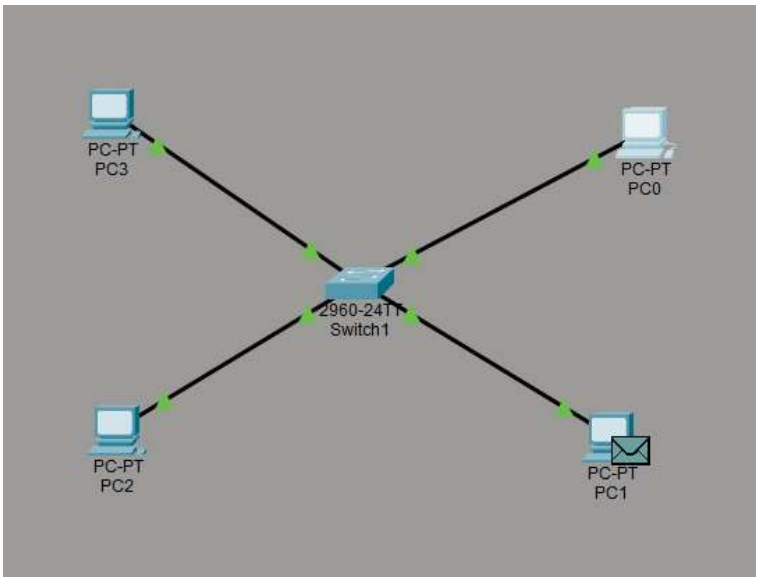

Fig 1c: Reply from destination PC — unicast delivery after MAC learning

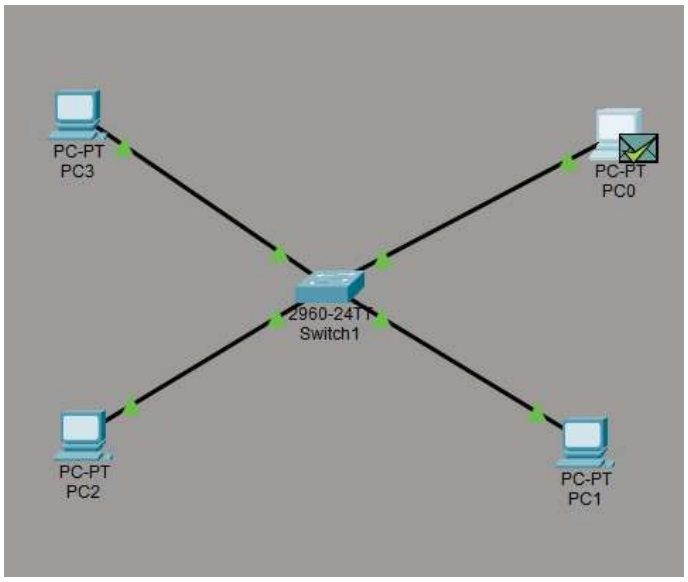

Fig 1d: Final state — switch forwards packets only to intended destination

Ping Results:

```
PC0 > ping 192.168.10.2
Reply from 192.168.10.2: bytes=32 time<1ms TTL=128 (x4)
Packets: Sent=4, Received=4, Lost=0 (0% loss) | Avg=0ms

PC0 > ping 192.168.10.3
Reply from 192.168.10.3: bytes=32 time<1ms TTL=128 (x4)
Packets: Sent=4, Received=4, Lost=0 (0% loss) | Avg=0ms

PC0 > ping 192.168.10.4
Reply from 192.168.10.4: bytes=32 time<1ms TTL=128 (x4)
Packets: Sent=4, Received=4, Lost=0 (0% loss) | Avg=0ms
```

Observation:

All four devices were able to communicate successfully through the switch without any packet loss. During the initial ping attempt, the switch broadcast the frame to all its ports because it had not yet learned the MAC addresses of the connected devices. Once the switch identified and stored the MAC address information in its MAC table, it started sending frames only through the specific port associated with the intended destination. This process, called selective unicast forwarding, minimizes unnecessary data transmission across the network. As a result, network efficiency improves significantly compared to a hub-based setup. The very low round-trip times, consistently under one millisecond, further demonstrate the high performance and reliability of a switch-based star topology.

Task 2: Star Topology (Hub)

The same star network setup was created again, but the switch was replaced with a Hub-PT (Hub0). Four PCs (PC0 to PC3) were connected to the hub and given IP addresses in the 192.168.10.x/24 range. The same ping tests were performed to compare the hub's performance with the switch-based star topology.

Network Topology Screenshots:

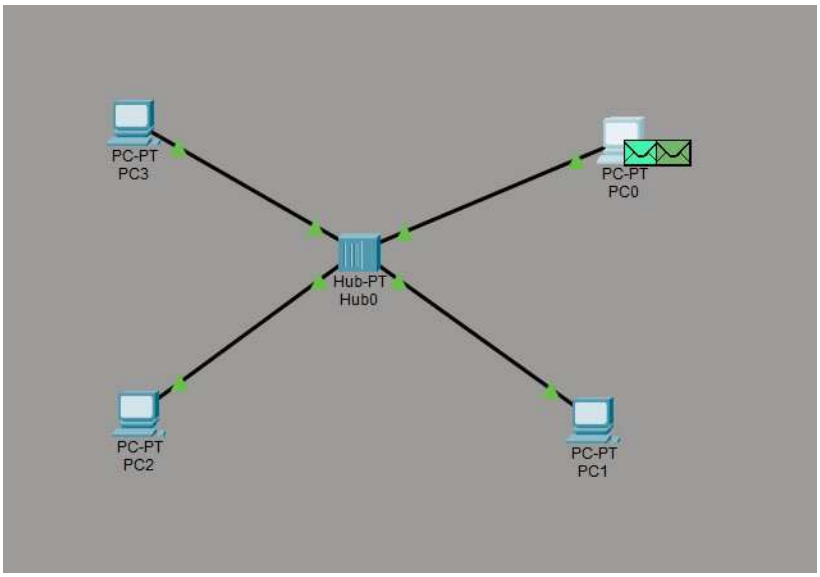

Fig 2a: PC0 sends ping — hub broadcasts frame to ALL connected devices

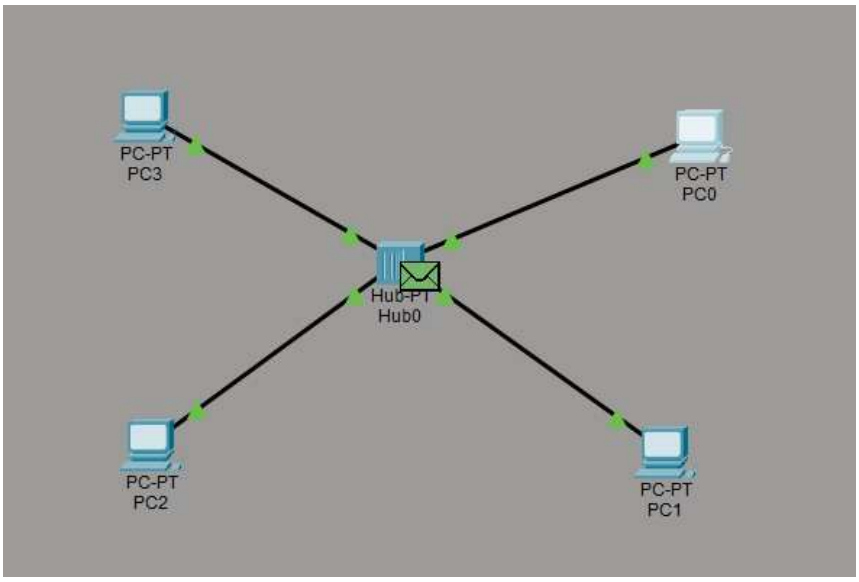

Fig 2b: Frame propagation — all devices receive the broadcast (including non-targets)

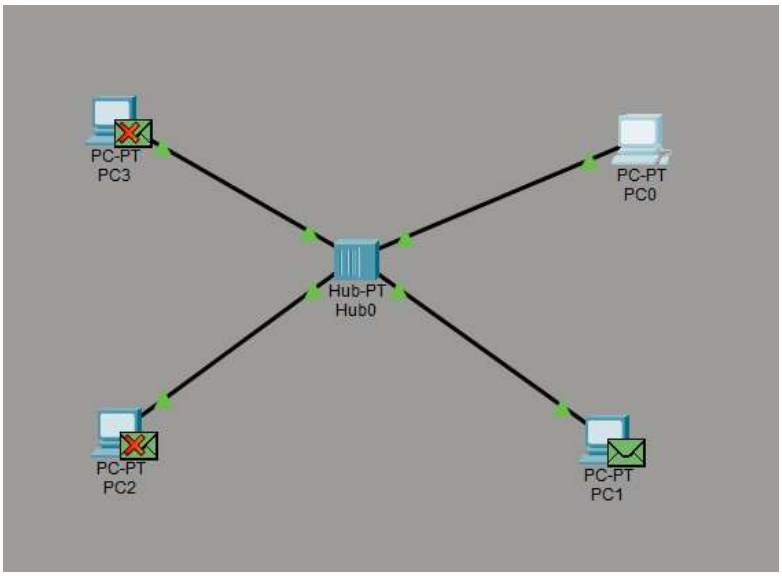

Fig 2c: Destination responds — reply also broadcast to all ports by hub

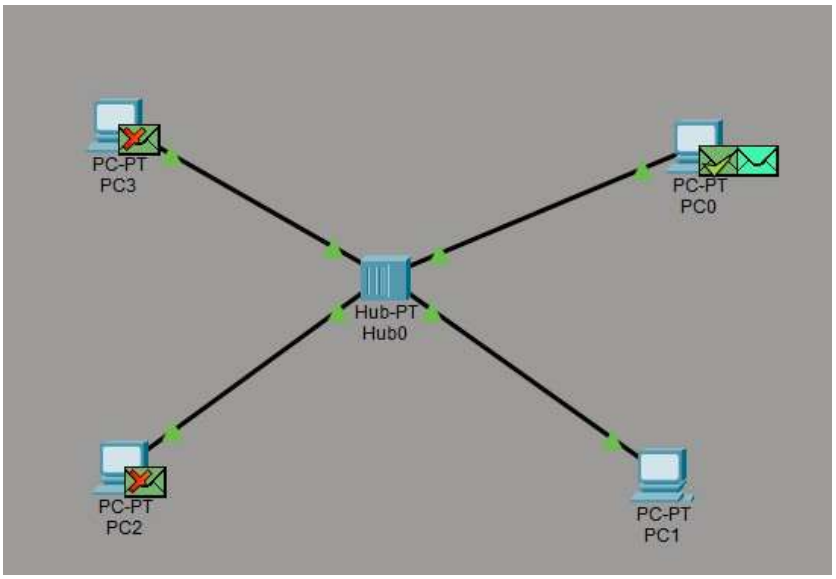

Fig 2d: Subsequent pings — hub continues to broadcast every frame (no MAC learning)

Ping Results:

```
PC0(1) > ping 192.168.10.6
Reply from 192.168.10.6: bytes=32 time<1ms TTL=128 (x4) Packets:
Sent=4, Received=4, Lost=0 (0% loss) | Max=1ms, Avg=0ms

PC0(1) > ping 192.168.10.7
Reply from 192.168.10.7: bytes=32 time<1ms TTL=128 (x4)
Packets: Sent=4, Received=4, Lost=0 (0% loss) | Avg=0ms

PC0(1) > ping 192.168.10.8
Reply from 192.168.10.8: bytes=32 time<1ms TTL=128 (x4) Packets:
Sent=4, Received=4, Lost=0 (0% loss) | Max=1ms, Avg=0ms
```

Observation:

All devices were able to communicate through the hub without any packet loss. However, unlike a switch, the hub sends every frame to all connected ports, not just the destination device. It does not have a MAC address table and cannot learn device addresses. Because of this, every device receives and checks every frame, even if it is not meant for it, which wastes network bandwidth and system resources. In larger networks, this can lead to network congestion and more collisions, since all devices share the same collision domain. Therefore, a switch-based topology is more efficient and better suited for real-world networks.

Task 3: Ring-like Topology (Switch Loop with STP)

Three 2960-24TT switches (Switch0, Switch1, and Switch2) were connected in a triangular shape to create a ring-like topology. Six PCs (PC0 to PC5) were connected across these switches. The extra links between the switches created multiple paths, which could cause broadcast storms if not controlled. To prevent this, Spanning Tree Protocol (STP) was enabled. STP automatically blocked one redundant link, keeping the network loop-free while still providing a backup path in case of failure.

Network Topology Screenshots:

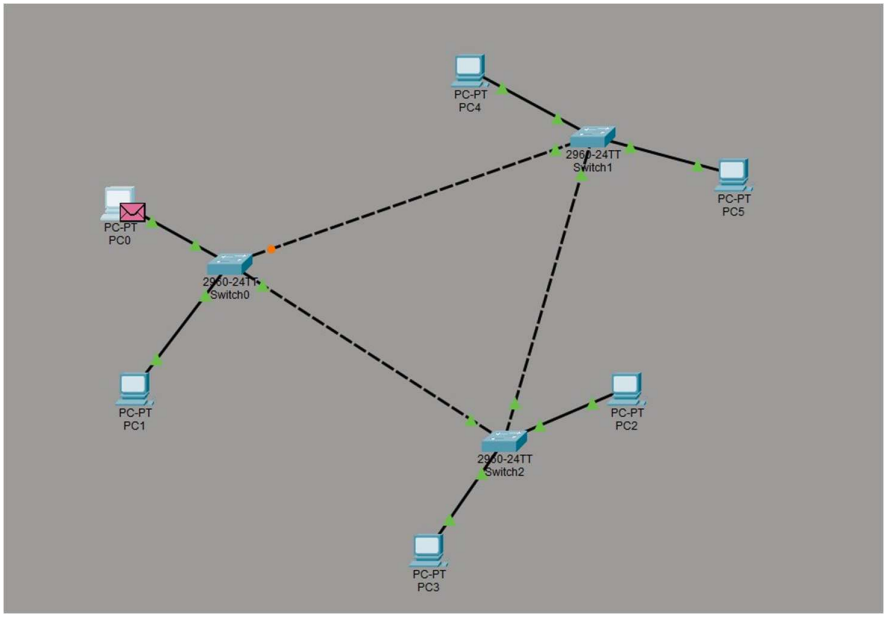

Fig 3a: PC0 initiates ping — STP has already blocked redundant links (shown as dashed)

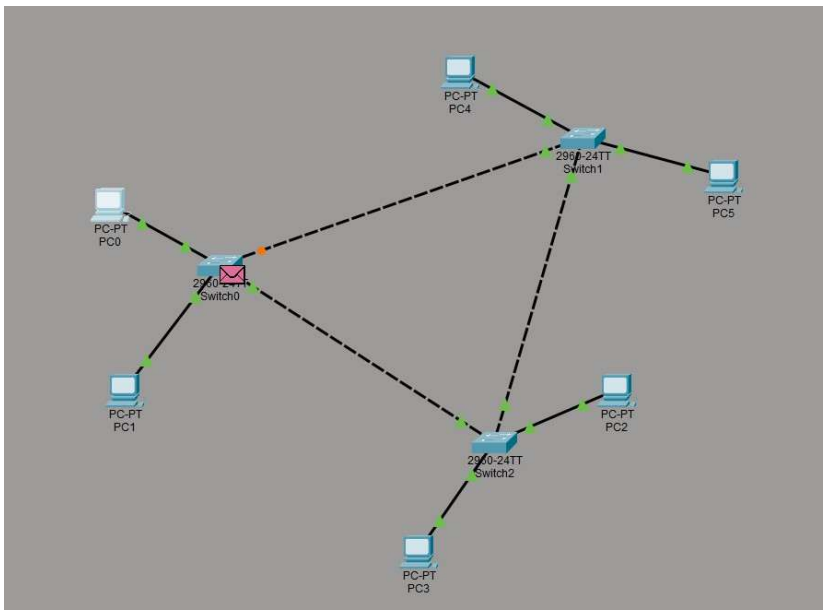

Fig 3b: Frame traverses active path through Switch0 → Switch2

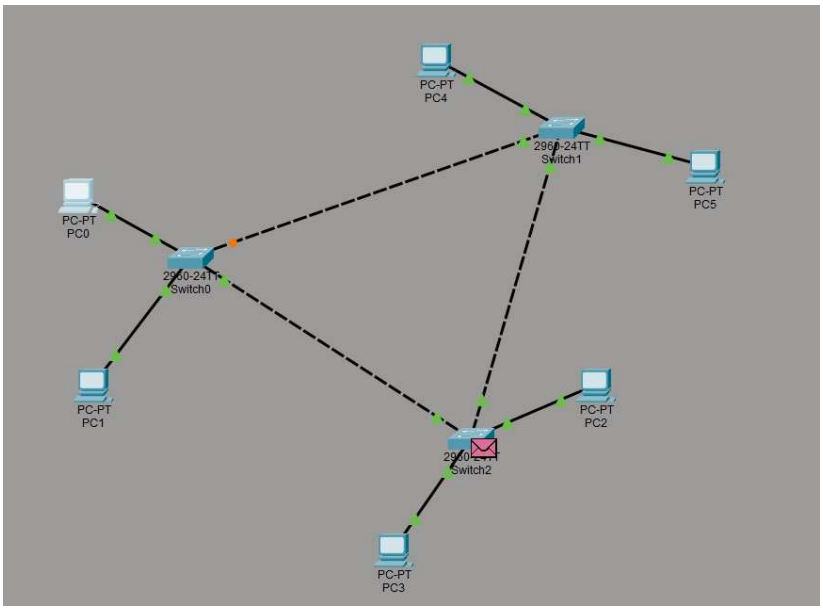

Fig 3c: Frame reaches Switch2, forwarded to destination PC

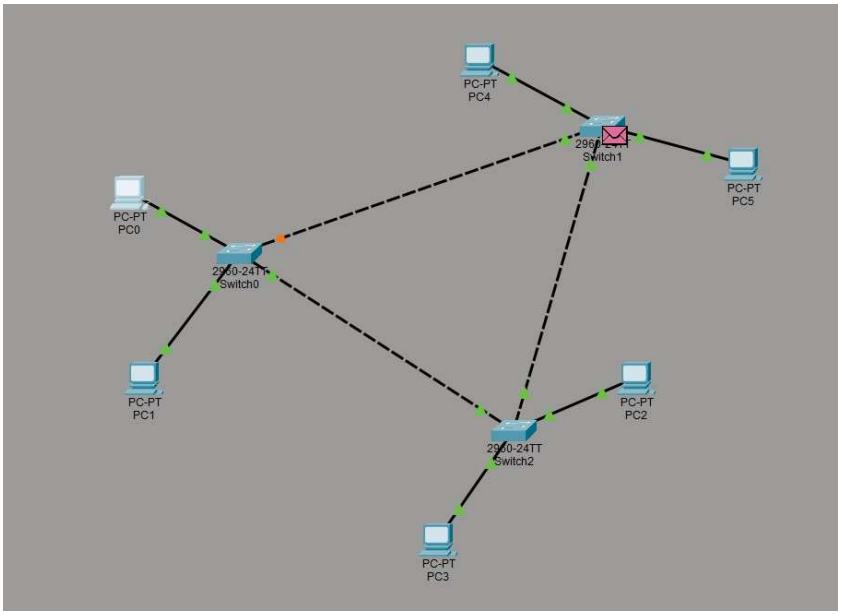

Fig 3d: Reply packet travelling back along the STP-selected active path

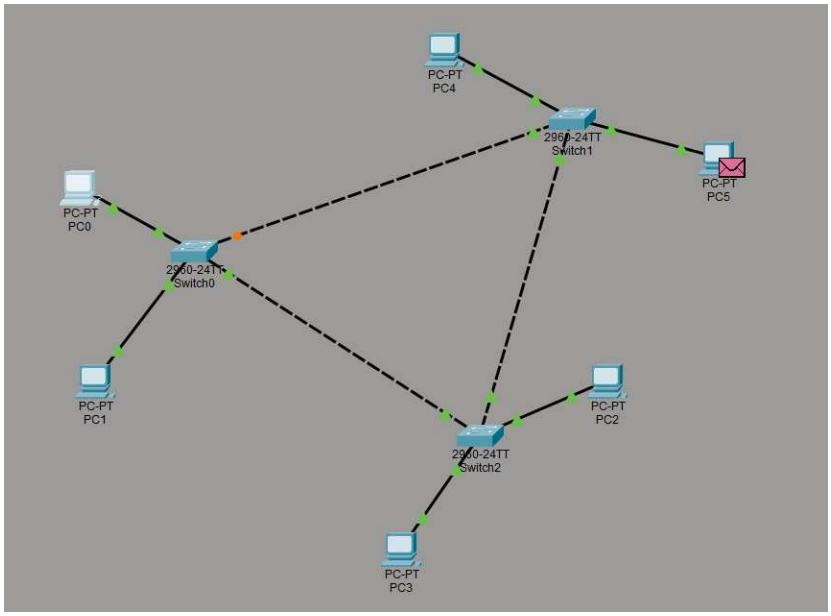

Fig 3e: Ping to second destination — traffic routed across multiple switches

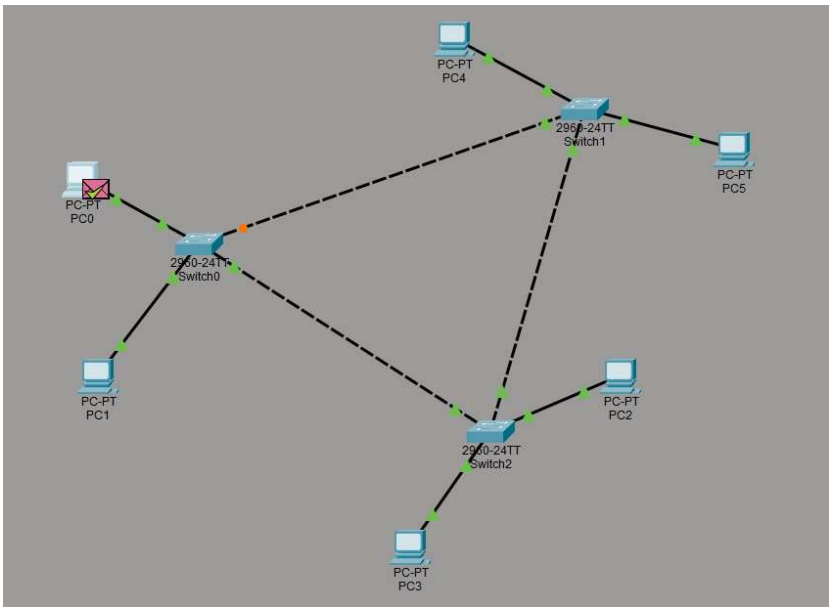

Fig 3f: Network stable — STP maintains loop-free topology throughout

Ping Results:

```
PC0(2) > ping 192.168.10.212
Reply from 192.168.10.212: bytes=32 time<1ms TTL=128 (x4)
Packets: Sent=4, Received=4, Lost=0 (0% loss) | Avg=0ms

PC0(2) > ping 192.168.10.211
Reply from 192.168.10.211: bytes=32 time<1ms TTL=128 (x4)
Packets: Sent=4, Received=4, Lost=0 (0% loss) | Max=1ms, Avg=0ms
```

Observation:

All devices communicated successfully across the switches with zero packet loss. STP automatically selected a root bridge and blocked one redundant link to prevent loops (shown as dashed lines in the screenshots). This prevented broadcast storms while still keeping a backup path available. The network remained stable, and no loops were formed. Data was sent through the active links chosen by STP, and all ping tests showed very fast response times under one millisecond.

Task 4: Failure Test (Ring Topology)

To test fault tolerance, one active link in the ring topology was intentionally disconnected. This simulated a real-world network failure, such as a cable or switch port failure. STP detected the change, unblocked the previously blocked link, and created a new loop-free path automatically. This allowed the network to restore connectivity without any manual configuration.

Network Topology Screenshots:

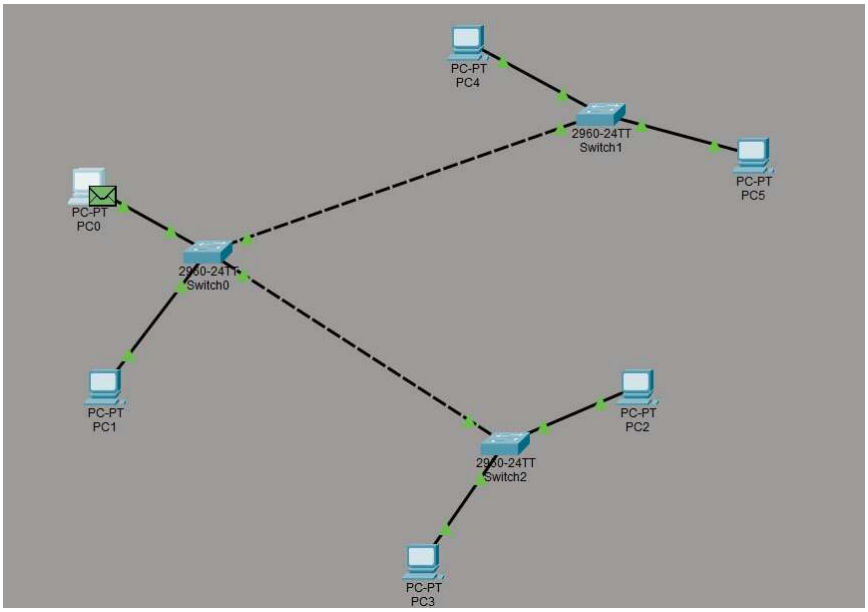

Fig 4a: Active link disconnected — STP detects topology change, begins recalculation

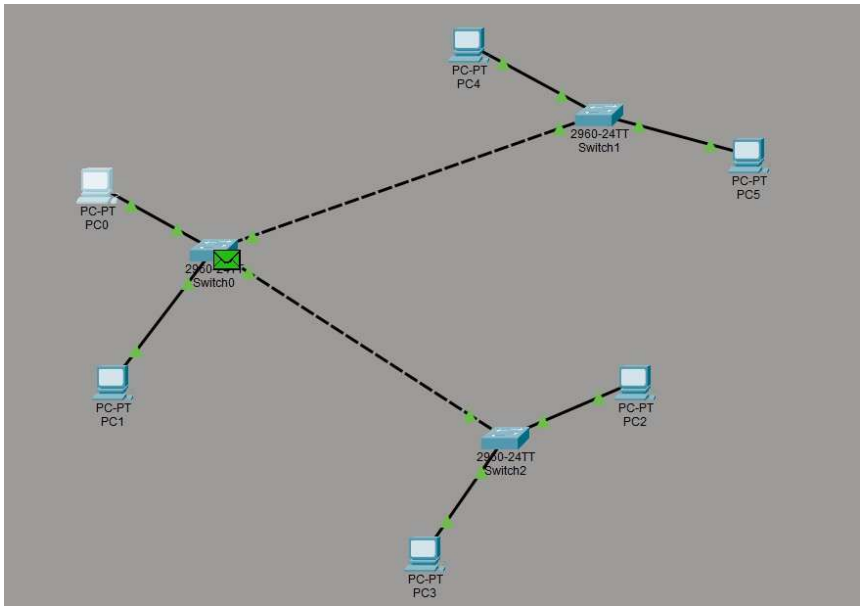

Fig 4b: STP activates previously blocked port to restore connectivity

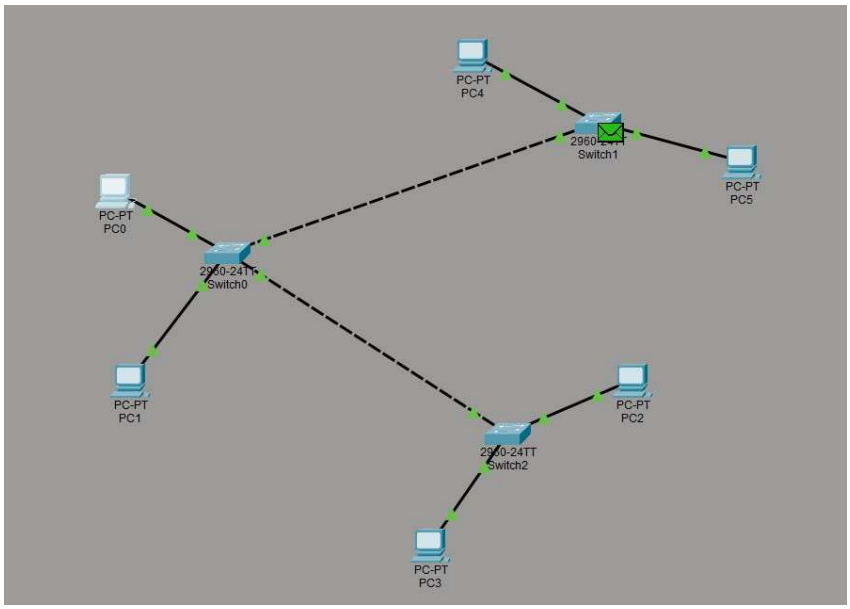

Fig 4c: New active path established — ping test initiated to verify connectivity

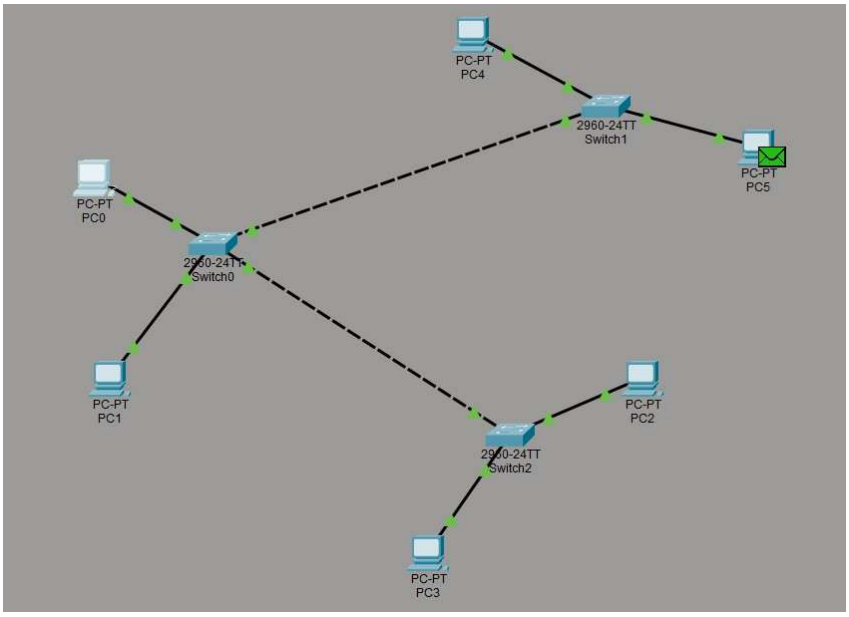

Fig 4d: Reply received — communication restored via alternate path

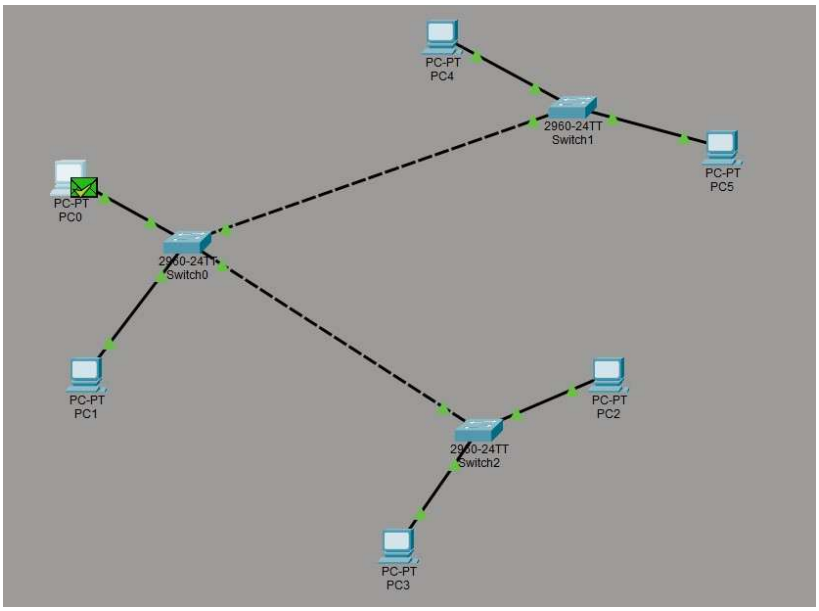

Fig 4e: Network fully stable on alternate path — zero packet loss confirmed

Ping Results:

```
PC0(3) > ping 192.168.10.218
Reply from 192.168.10.218: bytes=32 time<1ms TTL=128 (x4)
Packets: Sent=4, Received=4, Lost=0 (0% loss) | Avg=0ms

PC0(3) > ping 192.168.10.217
Reply from 192.168.10.217: bytes=32 time<1ms TTL=128 (x4)
Packets: Sent=4, Received=4, Lost=0 (0% loss) | Max=2ms, Avg=0ms
```

Observation:

After disconnecting the link, STP automatically recalculated the network and activated the previously blocked redundant port. This restored full communication within the STP convergence time. All ping tests were successful with zero packet loss, confirming that the ring topology can handle failures effectively. A slightly higher maximum round-trip time of 2 ms was observed in one test, which was due to the short delay during STP recalculation. This shows that a ring topology with STP is more reliable and fault tolerant than a simple star topology, where a single link failure can disconnect a device permanently.

Conclusion

Topology	Device	Packet Loss	Fault Tolerance	Efficiency
Star	Switch	0%	Low	High (unicast)
Star	Hub	0%	Low	Low (broadcast always)
Ring-like + STP	Switch	0%	Medium	High
Ring Failure Test	Switch	0%	High	High (auto-recovery)

This assignment provided practical experience with four basic network topologies. The switch-based star topology proved to be the most suitable for everyday LAN use because it forwards data intelligently and is easy to manage. The hub-based star topology worked correctly in a small setup but showed clear inefficiencies, making it unsuitable for modern networks. The ring-like topology using STP demonstrated how extra links can provide backup paths while still preventing network loops. The failure test showed that STP can automatically recover the network after a link failure with very little disruption, making this topology reliable and useful in environments where continuous network availability is important.